

ت (6)

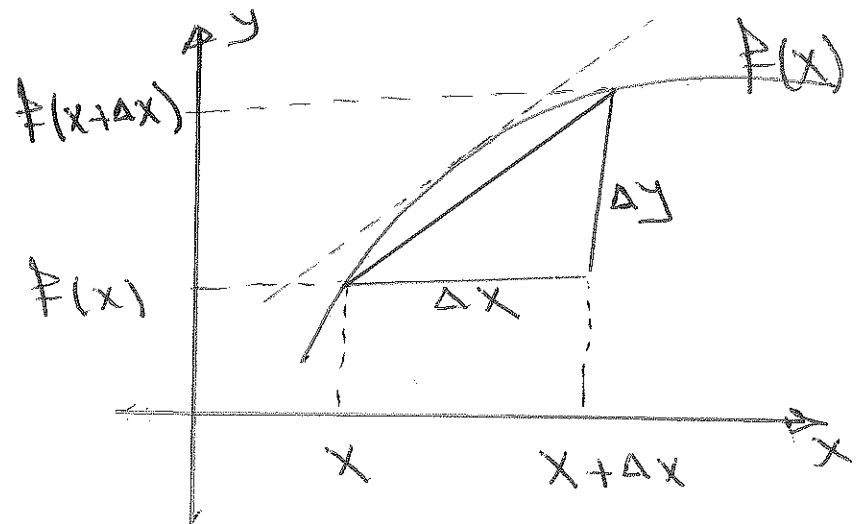
* Rules of Differentiation
قواعد التفاضل

* Table of Differentiation
جدول التفاضل

* Implicit Differentiation
التفاضل الضمني

* The Differentiation :

التفاضل



في المستقيم

$$\text{Slope of Line} = \frac{\Delta y}{\Delta x} = \frac{f(x+\Delta x) - f(x)}{\Delta x}$$

في المماس

$$\text{Slope of tangent} = \text{Slope of Line} \quad (\Delta x \rightarrow 0)$$

$$\therefore \text{Slope of tangent} = \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$$

$$f'(x) = \text{Slope of tangent} = \text{Derivative}$$

= rate of change

معدل التغير

مثال : كلما زادت قيمة المتغير $\frac{dy}{dx} = 5$ ←

بوحدة واحدة كلما زاد قيمة

المتغير (5) خمسة وحدات

note

$$y \equiv f(x)$$

دالة y نعتبر x دالة من

المشتق الأول
1st derivative:

$$\frac{dy}{dx} \equiv \frac{df(x)}{dx} \equiv y' \equiv f'(x)$$

المشتق الثاني
2nd derivative:

$$\frac{d^2y}{dx^2} \equiv \frac{d^2f(x)}{dx^2} \equiv y'' \equiv f''(x)$$

المشتق الثالث
3rd derivative:

$$\frac{d^3y}{dx^3} \equiv \frac{d^3f(x)}{dx^3} \equiv y''' \equiv f'''(x)$$

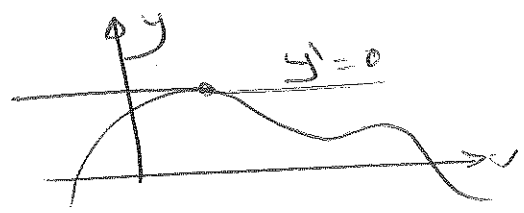
المشتق الرابع
4th derivative:

$$\frac{d^4y}{dx^4} \equiv y^{(4)}$$

$$\frac{d^7y}{dx^7} \equiv y^{(7)}$$

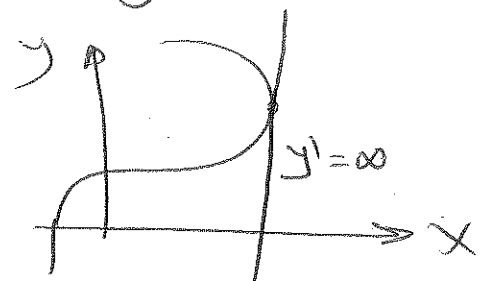
note

If $y' = 0$ $\rightarrow$ Horizontal tangent its equation $y = \text{Constant}$



If $y' = \infty$ $\rightarrow$ Vertical tangent

its equation $x = \text{Constant}$



→ Table of differentiation: جدول التفاضل

تفاضل

تفاضل

$f(x)$	$f'(x)$	$f(x)$	$f'(x)$
Constant	Zero	x^n	$n x^{n-1}$
$\sqrt{u}$	$\frac{1}{2\sqrt{u}} \cdot u'$	$\frac{1}{u}$	$-\frac{1}{u^2} \cdot u'$
e^u	$e^u \cdot u'$	a^u	$a^u \cdot \ln a \cdot u'$
$\ln u$	$\frac{1}{u} \cdot u'$	$\log_a u$	$\frac{1}{u} \cdot \frac{1}{\ln a} \cdot u'$
$\sin u$	$\cos u \cdot u'$	$\cos u$	$-\sin u \cdot u'$
$\tan u$	$\sec^2 u \cdot u'$	$\cot u$	$-\operatorname{cosec}^2 u \cdot u'$
$\sec u$	$\sec u \cdot \tan u \cdot u'$	$\operatorname{cosec} u$	$-\operatorname{cosec} u \cdot \cot u \cdot u'$
$\sin^{-1} u$	$\frac{1}{\sqrt{1-u^2}} \cdot u'$	$\cos^{-1} u$	$\frac{-1}{\sqrt{1-u^2}} \cdot u'$
$\tan^{-1} u$	$\frac{1}{1+u^2} \cdot u'$	$\cot^{-1} u$	$\frac{-1}{1+u^2} \cdot u'$
$\sec^{-1} u$	$\frac{1}{u\sqrt{u^2-1}} \cdot u'$	$\operatorname{cosec}^{-1} u$	$\frac{-1}{u\sqrt{u^2-1}} \cdot u'$
$\sinh u$	$\cosh u \cdot u'$	$\cosh u$	$\sinh u \cdot u'$
$\tanh u$	$\operatorname{sech}^2 u \cdot u'$	$\coth u$	$-\operatorname{cosech}^2 u \cdot u'$
$\operatorname{sech} u$	$-\operatorname{sech} u \cdot \tanh u \cdot u'$	$\operatorname{cosech} u$	$-\operatorname{cosech} u \cdot \coth u \cdot u'$
$\sinh^{-1} u$	$\frac{1}{\sqrt{1+u^2}} \cdot u'$	$\cosh^{-1} u$	$\frac{1}{\sqrt{u^2-1}} \cdot u'$
$\tanh^{-1} u$	$\frac{1}{1-u^2} \cdot u'$		

اللمظة: الدالة (u) في الجدول تمثل أي دالة ص (x)

→ Rules of Differentiation:

قواعد التفاضل

$$1) \frac{d}{dx} (C \times f(x)) = C \frac{d}{dx} (f(x))$$

$$2) \frac{d}{dx} (f(x) \pm g(x)) = \frac{d}{dx} (f(x)) \pm \frac{d}{dx} (g(x))$$

$$3) \frac{d}{dx} (f(x) \cdot g(x)) = f(x) \cdot \frac{d}{dx} (g(x)) + g(x) \cdot \frac{d}{dx} (f(x))$$

$$4) \frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \frac{g(x) \cdot \frac{d}{dx} (f(x)) - f(x) \cdot \frac{d}{dx} (g(x))}{(g(x))^2}$$

→ Chain Rule:

القاعدة السلسلة

$$\frac{d}{dx} [f(g(x))] = f'(g(x)) \times g'(x)$$

تفاضل دالة باخر الدالة

→ Function power function:

دالة القوة

$$\frac{d}{dx} [f(x)^{g(x)}] = g(x) [f(x)]^{g(x)-1} \cdot f'(x) + [f(x)]^{g(x)} \cdot \ln(f(x)) \cdot g'(x)$$

القاعدة العامة للتفاضل

للملاحظة: دائماً في التفاضل نستعمل هذا الترتيب

Power $\rightsquigarrow$ Function $\rightsquigarrow$ angle

يعني نفاضل هندس دئول ثم نفاضل الدالة ثم نفاضل الزاوية

ex: $y = \sin^3 \sqrt{x}$

$$y' = \underbrace{3 \sin^2 \sqrt{x}}_{\text{تفاضل الهندس}} \cdot \underbrace{\cos \sqrt{x}}_{\substack{\text{تفاضل الدالة} \\ \text{بدون أثر على}}}} \cdot \underbrace{\frac{1}{2\sqrt{x}}}_{\text{تفاضل الزاوية}}$$

ex: $y = \ln^5 \left(\tan \frac{1}{x} \right)$

$$y' = \underbrace{5 \ln^4 \left(\tan \frac{1}{x} \right)}_{\text{تفاضل هندس}} \cdot \underbrace{\frac{1}{\tan \frac{1}{x}}}_{\substack{\text{تفاضل الدالة} \\ \text{اللوغاريتمية في } \ln}} \cdot \underbrace{\sec^2 \frac{1}{x}}_{\substack{\text{تفاضل } \ln \\ \text{بدل في } \ln}} \cdot \underbrace{\frac{-1}{x^2}}_{\substack{\text{تفاضل} \\ \text{حاصل في} \\ \text{الدالة المقلبة}}}}$$

للملاحظة: لو الدالة من صيغة $\tan^{-1}$ علينا اختصارها مباشرة
نعمل الاختصار قبل التفاضل

Ex: $y = \tan^{-1}(6x^3)$ Find $\frac{dy}{dx}$

Soln

$$y = \tan^{-1} \left[\tan \left(\frac{\pi}{2} - x^3 \right) \right] \quad \therefore y = \frac{\pi}{2} - x^3$$

$$\therefore \left| \frac{dy}{dx} \right| = -3x^2$$

Ex: $y = \cosh\left(\frac{1}{\ln(\sqrt{x} + e^{x^2})}\right)$

[6]

$$\therefore y' = \sinh\left(\frac{1}{\ln(\sqrt{x} + e^{x^2})}\right) \cdot \frac{-1}{\left(\ln(\sqrt{x} + e^{x^2})\right)^2} \cdot \frac{1}{\sqrt{x} + e^{x^2}} \cdot \left(\frac{1}{2\sqrt{x}} + 2x e^{x^2}\right)$$

Ex: $y = \tan^{-1}(\sqrt{\sinh x^2})$

$$y' = \frac{1}{1 + \sinh x^2} \cdot \frac{1}{2\sqrt{\sinh x^2}} \cdot \frac{1 \cdot 2x}{\sqrt{1+x^4}}$$

Ex: $y = (\tan^{-1} x)^{x^2}$

$$\therefore y' = x^2 (\tan^{-1} x)^{x^2-1} \cdot \frac{1}{1+x^2}$$

$$+ (\tan^{-1} x)^{x^2} \cdot \frac{1}{\ln \tan^{-1} x} \cdot 2x$$

Exe $y = (\tanh x)^{\tanh x} + \sinh^{-1}(3^x)$

[7]

$$y' = \tanh x (\tanh x)^{\tanh x - 1} \cdot \frac{1}{x} \\ + (\tanh x)^{\tanh x} \cdot \ln(\tanh x) \cdot \operatorname{sech}^2 x \\ + \frac{1}{\sqrt{1+(3^x)^2}} \cdot 3^x \cdot \ln 3.$$

Ex: $y = 3^{\operatorname{sech} x^2} + (\sqrt{x})^{\coth^{-1} x^3}$

$$y' = 3^{\operatorname{sech} x^2} \cdot -\operatorname{sech} x^2 \tanh x^2 \cdot 2x \cdot \ln 3 \\ + \coth^{-1} x^3 \cdot (\sqrt{x})^{\coth^{-1} x^3 - 1} \cdot \frac{1}{2\sqrt{x}} \\ + (\sqrt{x})^{\coth^{-1} x^3} \cdot \ln(\sqrt{x}) \cdot \frac{1}{\sqrt{x^6 - 1}} \cdot 3x^2$$

Ex: $y = \sqrt{x} \tanh^{-1}(\sin x)$

[8]

$$\therefore y' = \sqrt{x} \cdot \frac{1}{1 - \sin^2 x} \cdot \cos x + \frac{1}{2\sqrt{x}} \tanh^{-1}(\sin x)$$

Ex: $y = \sec^3(\tan^2 \frac{1}{x})$

$$\therefore y' = 3 \sec^2(\tan^2 \frac{1}{x}) \cdot \sec(\tan^2 \frac{1}{x}) \cdot \tan(\tan^2 \frac{1}{x}) \cdot 2 \tan \frac{1}{x} \cdot \sec^2 \frac{1}{x} \cdot \frac{-1}{x^2}$$

Ex: $y = \cos^2 \sqrt{x} \cdot \sec^{-1} x^2$

$$\therefore y' = \cos^2 \sqrt{x} \cdot \frac{1}{x^2 \sqrt{x^4 - 1}} \cdot 2x + 2 \cos \sqrt{x} \cdot -\sin \sqrt{x} \cdot \frac{1}{2\sqrt{x}} \cdot \sec^{-1} x^2$$

Ex: $y = \tan^{-1}\left(\frac{1}{x^2}\right) + \cos^{-1} \frac{1}{\sqrt{x}}$

$$\therefore y' = \frac{1}{1 + \frac{1}{x^4}} \cdot -2x^{-3} + \frac{-1}{\sqrt{1 - \frac{1}{x}}} \cdot \frac{-1}{2} x^{-3/2}$$

Ex: $y = (\cos x^3)^{\sqrt[3]{x}} + (4)^{\cos^{-1} \frac{1}{x}}$

$$\begin{aligned} \therefore y' &= \sqrt[3]{x} (\cos x^3)^{\sqrt[3]{x}-1} \cdot -\cos x^3 \cot x^3 \cdot 3x^2 \\ &+ (\cos x^3)^{\sqrt[3]{x}} \cdot \ln \cos x^3 \cdot \frac{1}{3} x^{-\frac{2}{3}} \\ &+ (4)^{\cos^{-1} \frac{1}{x}} \cdot \ln 4 \cdot \frac{-1}{\frac{1}{x} \sqrt{\frac{1}{x^2}-1}} \cdot \frac{-1}{x^2} \end{aligned}$$

Ex: $y = \frac{x + \sinh(3x)}{3 + 2 \sin^{-1} \sqrt{x}}$

$$\therefore y' = \frac{(3 + 2 \sin^{-1} \sqrt{x})(1 + \cosh(3x) \cdot 3) - (x + \sinh(3x)) \cdot \frac{2}{\sqrt{1-x}} \cdot \frac{1}{2\sqrt{x}}}{(3 + 2 \sin^{-1} \sqrt{x})^2}$$

Note

$$\sqrt[3]{x} = x^{1/3} \xrightarrow{\text{تفاضل}} \frac{1}{3} x^{-2/3}$$

$$\sqrt{x^5} = x^{5/2} \xrightarrow{\text{تفاضل}} \frac{5}{2} x^{3/2}$$

$$\frac{1}{x^2} = x^{-2} \xrightarrow{\text{تفاضل}} -2 x^{-3}$$

$$\frac{4}{x^3} = 4 x^{-3} \xrightarrow{\text{تفاضل}} -12 x^{-4}$$

Ex: $y = \ln(x e^{\tan^{-1} x^2})$ Find y'

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Soln $\ln$ کے لیے e کی پاور کو باہر لے کر آؤ

$$\therefore y = \ln x + \ln e^{\tan^{-1} x^2} = \ln x + \tan^{-1} x^2$$

$$\therefore y' = \frac{1}{x} + \frac{1}{1+x^2} \cdot 2x$$

Ex: $y = \sec^{-1}(\operatorname{cosec} x^2)$

Find $\frac{dy}{dx}$ in simplest form سب سے سادہ

Soln,

$$y' = \frac{1}{\operatorname{cosec} x^2 \sqrt{\operatorname{cosec}^2 x^2 - 1}} \cdot -\operatorname{cosec} x^2 \cot x^2 \cdot 2x$$

$$= \frac{1}{\cancel{\operatorname{cosec} x^2} \cdot \cancel{\cot x^2}} \cdot -\cancel{\operatorname{cosec} x^2} \cancel{\cot x^2} \cdot 2x$$

$$\therefore y' = -2x$$

note

$$\operatorname{cosec}^2 \theta - 1 = \cot^2 \theta$$

$$\sqrt{\operatorname{cosec}^2 \theta - 1} = \cot \theta$$

Ex: $y = \sinh^{-1}(\tan x)$

Find y'' in simplest form

Soln

$$y' = \frac{1}{\sqrt{1 + \tan^2 x}} \cdot \sec^2 x$$

$$= \frac{1}{\sqrt{\sec^2 x}} \cdot \sec^2 x$$

$$\therefore y' = \sec x$$

$$\therefore y'' = \sec x \tan x$$

Ex: $y = \frac{1 - e^{-2\sqrt{x}}}{1 + e^{-2\sqrt{x}}}$

Find y'

Soln

هنا تعريف (y) مع شكل تعريف tanh

فلا نفضل قبل التفاضل كتابته بهذه

$$\tanh 0 = \frac{e^0 - e^{-0}}{e^0 + e^{-0}} = \frac{e^{20} - 1}{e^{20} + 1} = \frac{1 - e^{-20}}{1 + e^{-20}}$$

$$\therefore y = \frac{1 - e^{-2\sqrt{x}}}{1 + e^{-2\sqrt{x}}} = \tanh \sqrt{x}$$

$$\therefore y' = \operatorname{sech}^2 \sqrt{x} \cdot \frac{1}{2\sqrt{x}}$$

Ex: $y = \cos(\sin^{-1} x^2) + \tan(\sec^{-1} x)$

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Find y' in simplest form.

Soln

$$y' = -\sin(\sin^{-1} x^2) \cdot \frac{1}{\sqrt{1-x^4}} \cdot 2x + \sec^2(\sec^{-1} x) \cdot \frac{1}{x\sqrt{x^2-1}}$$

$$\therefore y' = \frac{-2x^3}{\sqrt{1-x^4}} + x^2 \cdot \frac{1}{x\sqrt{x^2-1}}$$

$$y' = \frac{-2x^3}{\sqrt{1-x^4}} + \frac{x}{\sqrt{x^2-1}}$$

حل آخر

نستخدم الدالة (y) من جدول

$$y = \cos(\sin^{-1} x^2) + \tan(\sec^{-1} x)$$

$$\therefore y = \sqrt{1 - \sin^2(\sin^{-1} x^2)} + \sqrt{\sec^2(\sec^{-1} x) - 1}$$

$$\therefore y = \sqrt{1 - x^4} + \sqrt{x^2 - 1}$$

أفكر في الدالة
من الجدول!

أبلى

$$\therefore y' = \frac{-4x^3}{2\sqrt{1-x^4}} + \frac{2x}{2\sqrt{x^2-1}}$$

Ex: $y = \ln(\sec x + \tan x)^6$

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Find y'' in simplest form

Soln $y = 6 \ln(\sec x + \tan x)$

$$\therefore y' = 6 \frac{\sec x \tan x + \sec^2 x}{\sec x + \tan x} = 6 \frac{\sec x (\cancel{\tan x} + \sec x)}{(\cancel{\sec x} + \cancel{\tan x})}$$

$$\therefore y' = 6 \sec x$$

$$\therefore y'' = 6 \sec x \tan x$$

Ex: $10 \sin x = e^{4y}$ Find $\frac{dy}{dx}$

Soln

write $\ln$ is $\ln$

$$\ln(10 \sin x) = 4y$$

$$\therefore y = \frac{1}{4} \ln(10 \sin x)$$

$$\therefore \frac{dy}{dx} = \frac{1}{4} \frac{10 \cos x}{10 \sin x} = \frac{1}{4} \cot x$$

Ex: $y = e^{\tan^{-1} 2 \ln x^2}$

13

Find y'' in simplest form

Soln

$$y = e^{\tan^{-1} 2 \ln x^2} = e^{2 \ln x^2} = x^4$$

$$\therefore y' = 4x^3$$

$$\therefore y'' = 12x^2$$

Ex: IF $\frac{d}{dx} F(x) = G(x)$

Find $\frac{d}{dx} F(2x^2) = \frac{d}{dx} \left(\frac{1}{F(x)} \right)$

Soln

$$\frac{d}{dx} F(2x^2) = 4x \cdot G(2x^2)$$

$$\frac{d}{dx} \left(\frac{1}{F(x)} \right) = \frac{-1}{(F(x))^2} \cdot G(x)$$

Ex: If $\frac{d}{dx} F(ax) = G(x)$

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Find $\frac{d}{dx} F(x)$, $\frac{d}{dx} F(2ax)$

Soln

$$\frac{d}{dx} F(ax) = G(x)$$

قسمنا ط با دض
الاول مع (a)

∴ قسم ط با دض G
(a) مع وقسم الناتج
مع (a) ايضاً

$$\therefore \frac{d}{dx} F(x) = \frac{1}{a} G\left(\frac{x}{a}\right)$$

$$\therefore \frac{d}{dx} F(2ax) = 2 G(2x) \leftarrow \begin{array}{l} \text{هنا ضربنا في 2} \\ \text{∴ ضربنا دض G في 2، وقسم الناتج في 2} \end{array}$$

Implicit differentiation:

التفاضل الخفي

$$\left. \begin{array}{l} y = f(x) \\ \text{ex: } y = x^2 + \tan^3 x \end{array} \right\} \Rightarrow \text{Direct function}$$

دالة صريحة
لا في حرف و x في حرف آخر

$$\left. \begin{array}{l} f(x, y) = 0 \\ \text{ex: } xy + \sin\left(\frac{x^2}{y}\right) + \sqrt{x+y} = 0 \end{array} \right\} \Rightarrow \text{Implicit function}$$

دالة خفية
x و y متداخلين ولا يمكن فصلهم

في الدوال الخفية نفاضل المعادلة كلياً بالنسبة إلى x
ثم نجمع (الـ y) في حرف لو حدها صفر لمعادلة

note

$$\frac{d}{dx} (xy) = xy' + y$$

حرف التفاضل بالنسبة إلى x

$$\frac{d}{dx} (y^3) = 3y^2 y'$$

$$\frac{d}{dx} (x^2 y^2) = x^2 \cdot 2y y' + 2x y^2$$

$$\frac{d}{dx} [x(y')^2] = x \cdot 2(y') \cdot y'' + (y')^2$$

Ex: $e^{xy} + \sin\left(\frac{y}{x}\right) = 4$ Find y'

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Soln

$$e^{xy} \cdot [xy' + y] + \cos\left(\frac{y}{x}\right) \cdot \frac{xy' - y}{x^2} = \text{zero}$$

$$xy' e^{xy} + ye^{xy} + \cos\left(\frac{y}{x}\right) \cdot \left[\frac{y'}{x} - \frac{y}{x^2} \right] = 0$$

$$xy' e^{xy} + ye^{xy} + \frac{y'}{x} \cos\left(\frac{y}{x}\right) - \frac{y}{x^2} \cos\left(\frac{y}{x}\right) = 0$$

$$y' \left[x e^{xy} + \frac{1}{x} \cos\left(\frac{y}{x}\right) \right] = \frac{y}{x^2} \cos\left(\frac{y}{x}\right) - ye^{xy}$$

$$y' = \frac{\frac{y}{x^2} \cos\left(\frac{y}{x}\right) - ye^{xy}}{x e^{xy} + \frac{1}{x} \cos\left(\frac{y}{x}\right)}$$

Ex: $x^2 y + \cos y^2 = \tanh \sqrt{y}$ Find y'

Soln $x^2 y' + 2xy - \sin y^2 \cdot 2yy' = \text{sech}^2 \sqrt{y} \cdot \frac{1}{2\sqrt{y}} y'$

$$\therefore y' \left[x^2 - 2y \sin y^2 - \operatorname{sech}^2 y \cdot \frac{1}{2\sqrt{y}} \right] = -2xy$$

$$\therefore y' = \frac{-2xy}{x^2 - 2y \sin y^2 - \operatorname{sech}^2 y \cdot \frac{1}{2\sqrt{y}}}$$

Exo $3x^2 + 2xy + y^2 = 2$

find y' at $(x=1)$

Soln $6x + 2xy' + 2y + 2yy' = 0$

$$\therefore y' (2x + 2y) = -6x - 2y$$

$$\therefore y' = \frac{-3x - y}{x + y}$$

at $\boxed{x=1}$ نقطة العبارة $x=1$
 حساب قيمة y $3 + 2y + y^2 = 2$

$$\therefore 1 + 2y + y^2 = 0$$

$$\therefore (1+y)^2 = 0$$

$$\therefore y = -1$$

$(1, -1)$ النقطة كما هو

$$\therefore y' \Big|_{(1,-1)} = \frac{-2}{0} = (\infty)$$

لاحظ معن $y' = \infty$

لكونه المماس عند هذه النقطة أفقي
ولو كان $y' = 0$ فمعنى ذلك أنه المماس أفقي

Ex: $e^y = x^2 + y^2$

Find equation of tangent Line to y at $(1,0)$

Soln
لدينا معادلة المماس عند النقطة $(1,0)$ فسنحسب y' عند $(1,0)$

$$\therefore e^y \cdot y' = 2x + 2yy'$$

$$\therefore y'(e^y - 2y) = 2x \quad \therefore y' = \frac{2x}{e^y - 2y}$$

$$\therefore y' \Big|_{(1,0)} = \frac{2}{1} = (2)$$

equation of tangent Line: $\frac{y-0}{x-1} = 2$

[note]

$$\therefore y = 2x - 2$$

معادلة أي مستقيم: $\frac{y-y_1}{x-x_1} = m$ slope

Ex: $f(x) = x^3 + 2x + \cos 2x$

Find $f^{(4)}(0)$

Soln

$f^{(4)}(0)$ $\xrightarrow{\text{مضاهيها}}$ المشتقة الرابعة عند $x=0$

$f(x) = x^3 + 2x + \cos 2x$

من الدرجة الثالثة لا تتفاضل
الرابع تكون صفر

$\therefore f^{(4)}(x) = (2)^4 \cos 2x$

كل مرة تفاضل تطلع (2)
الدالة بعد (4) تفاضلات لا تتغير

$\therefore f^{(4)}(0) = (2)^4 \cos(0)' = \boxed{16}$

Ex: $\ln y = xy$ Find y' at $x=0$

Soln

$\frac{1}{y} y' = xy' + y$

$$\text{If } y = xy \xrightarrow{\text{at } \boxed{x=0}} \text{If } y = 0 \Rightarrow \boxed{y=1}$$

$$\therefore y' \left(\frac{1}{y} - x \right) = y$$

$$\therefore y' = \frac{y}{\frac{1}{y} - x}$$

$$\therefore y' \Big|_{(0,1)} = \frac{1}{1} = \boxed{1}$$

Ex 8 $x^2 + axy - y^2 = 0$, $\frac{dy}{dx} = \frac{x}{y}$

Find the value of a

Soln

$$2x + axy' + ay - 2yy' = 0$$

$$y' [ax - 2y] = -2x - ay$$

$$\therefore y' = \frac{-2x - ay}{ax - 2y} = \frac{x}{y} \Rightarrow \therefore \boxed{a=0}$$

$$\therefore -2xy - ay^2 = ax^2 - 2xy$$

$$\therefore a(x^2 + y^2) = 0 \implies$$

$$x^2 + y^2 = \text{+ve not zero} \Rightarrow \boxed{a=0}$$